

Total No. of Questions : 6]

SEAT No. :

PE-540

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**S.E. (Electronics & Computer Eng./Electronics/E&TC/Electronics
(VLSI Design & Technology)/Electronics & Communication -
Advanced Communication Technology) (Insem)
DIGITAL CIRCUITS
(2019 Pattern) (Semester - I) (204182)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4. Q.5 or Q.6.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of non-programmable calculator is allowed.

- Q1)** a) List the various characteristics of logic family [6]
b) Explain the action of TTL NOT gate. [4]

OR

- Q2)** a) Draw and explain the working of 2-input TTL NAND gate. [6]
b) Write a note on logic families. [4]

- Q3)** a) What is combinational circuit? Give an example. [5]
b) Minimize the following expression using K-map and implement using logic gates: [5]
d () indicates don't care conditions
 $Y = \sum m(1, 3, 7, 11, 15) + \sum d(0, 2, 5)$

OR

- Q4)** a) Design 3-bit Binary to Gray code converter. [6]
b) Design full subtractor using half subtractors. [4]

- Q5)** a) Design and implement BCD to Excess-3 code converter using logic gates. [6]
b) Draw and explain the interfacing circuits for TTL with CMOS gates and CMOS with TTL. [4]

OR

- Q6)** a) Compare TTL and CMOS logic families. [4]
b) Minimize the following function using Quine Mc Clusky method. [6]
 $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$

